DATA ANALYTICS FOR BUSINESS

with Lizhen Xu, Ph.D.

Clustering: *K*-Means

Learning Objectives

- Explain the objective function of *K*-means clustering
- Describe *K*-means algorithm
- Explain why *K*-means algorithm works
- Choose the optimal *K*

What is Cluster Analysis?

- Finding *subgroups* within a data set
  - Observations within each group are quite *similar* to each other, or *homogeneous*
  - Observations in different groups are quite *different* from each other

- Unsupervised learning
  - Just to learn the structure, no outcome variables

- Cluster observations or variables
  - Cluster *observations* based on features/variables to find subgroups of obs.
  - Cluster *features/variables* based on observations to find subgroups of features

Applications of Cluster Analysis

- Market segmentation
- Industry analysis, market structure analysis
- Balanced portfolio
- Wide applications in various disciplines:
  - Genomics, biology, chemistry, medicine, astronomy, linguistics, …

K-Means Clustering

- ***K*-Means Clustering**
  - *K*: we need to *first specify the desired number of clusters K*
  - Means: algorithm based on the ***centroid*** of each cluster

- **Objective**
  - Assign each observation into one of the *K* clusters $\{C_1, C_2, \dots, C_K\}$
    - Clusters are *exhaustive*: each observation belongs to at least one cluster
    - Clusters are *exclusive* (*non-overlapping*): no obs. belongs to more than one cluster

K-Means Clustering (cont'd)

- **Objective Function**

$$\min_{C_1,\ldots,C_K} \left\{ \sum_{k=1}^{K} \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} \left(x_{ij} - x_{i'j}\right)^2 \right\}$$

  - Squared ***Euclidean distance***
  - Total Within-cluster Sum of Squares

- **Solving the Optimization Problem**
  - K^n different ways of partitioning
  - Heuristic algorithm to find a "pretty good" ***local optimum***

K-Means Algorithm

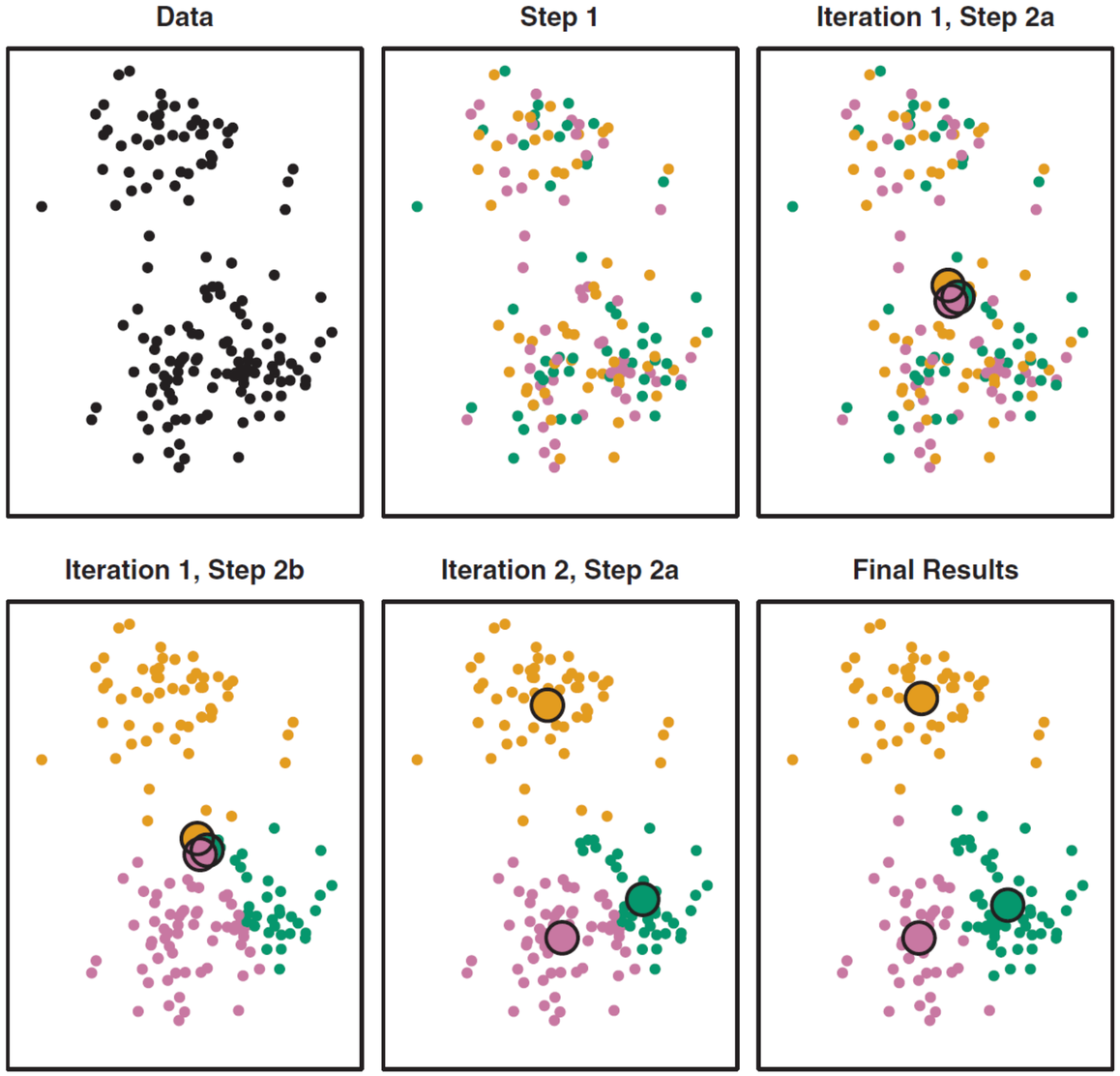
1. Randomly assign each observation into one of the *K* clusters
   - These are the initial cluster assignments

2. Iterate until the cluster assignments stop changing:

   a) For each of the *K* clusters, compute the cluster ***centroid***
      - A cluster ***centroid*** is a vector of means of variables for all the observations in that cluster
      - $\bar{x}_k = (\bar{x}_{k1}, \dots, \bar{x}_{kp})$, where $\bar{x}_{kj} = \frac{1}{|C_k|}\sum_{i \in C_k} x_{ij}$

   b) Assign each observation to the cluster whose centroid is closest
      - *Closest* defined by **Euclidean distance**

How *K*-Means Algorithm Works



James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning: with applications in R*. Springer. 2nd Ed. p 520. Retrieved March 29, 2022 from, https://hastie.su.domains/ISLR2/ISLRv2_website.pdf.

Why the Algorithm Works

- Recall the objective function

$$\min_{C_1,\ldots,C_K}\left\{\sum_{k=1}^{K}\frac{1}{|C_k|}\sum_{i,i'\in C_k}\sum_{j=1}^{p}\left(x_{ij}-x_{i'j}\right)^2\right\} \qquad (1)$$

- Note that it can be shown:

$$\frac{1}{|C_k|}\sum_{i,i'\in C_k}\sum_{j=1}^{p}\left(x_{ij}-x_{i'j}\right)^2 = 2\sum_{i\in C_k}\sum_{j=1}^{p}\left(x_{ij}-\bar{x}_{kj}\right)^2 \qquad (2)$$

- In each iteration, the algorithm reduces the objective function (1)
  - Given the centroids in the previous step, Step 2(b) reduces RHS of Eq.(2)
  - The new within-cluster SS, based on the new centroids in Step 2(a), is even lower

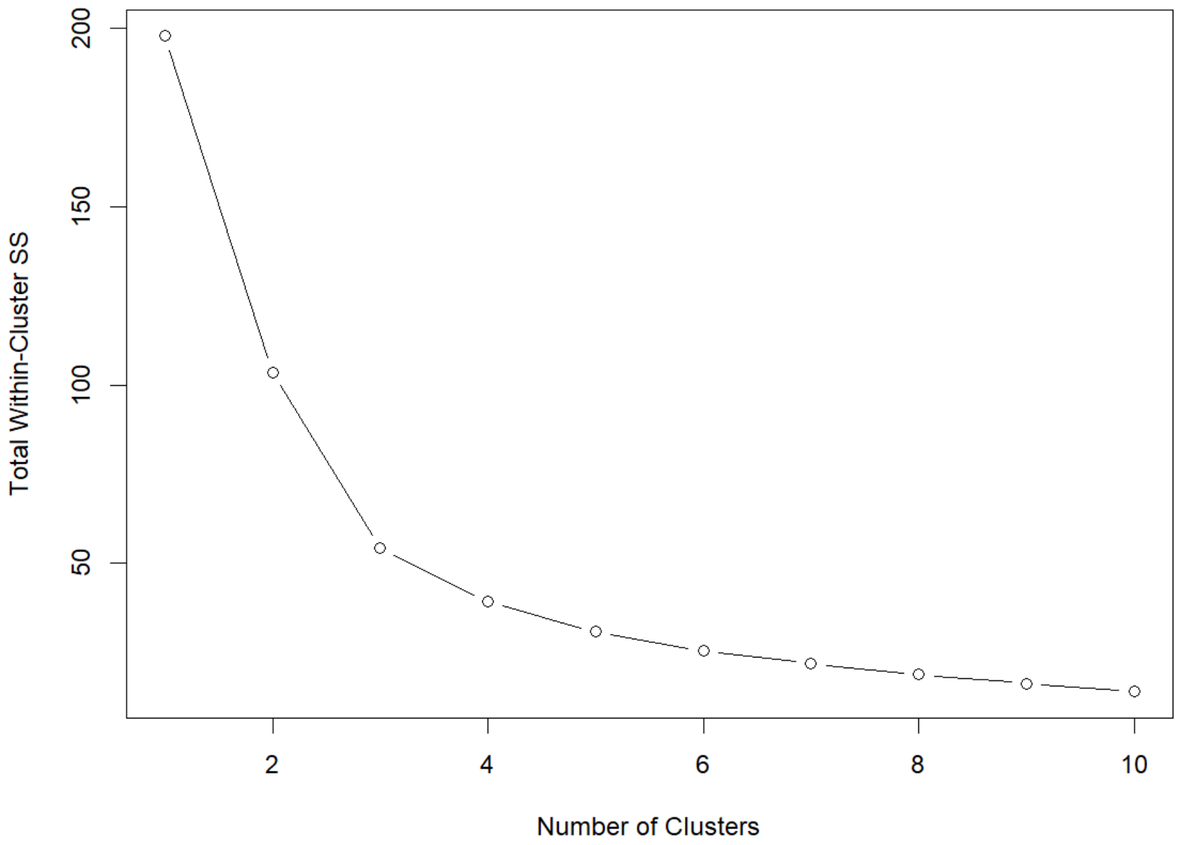
Why the Algorithm Works (cont'd)

- Find a ***local optimum***
  - Greedy algorithm

- Outcome depends on initial (random) assignments
  - Try different initial assignments
  - Pick the one with the ***smallest*** total within-cluster SS

How to Choose *K*

- Domain knowledge
- Practical consideration
- "Elbow Chart"
  - A line chart depicting the total within-cluster sum of squares for different choices of number of clusters *K*
  - Objective function's value always *decreases* as *K* increases
  - Pick the optimal *K* value around the *elbow* point



Summary

- *K*-means clustering: objective
- *K*-means algorithm
- Why *K*-means algorithm works
- How to choose the optimal *K*

DATA ANALYTICS FOR BUSINESS

with Lizhen Xu, Ph.D.

Clustering: Hierarchical

Learning Objectives

- Describe hierarchical clustering algorithm
- Interpret dendrograms
- Measure distances
  - Between observations
  - Between clusters
- Deal with practical issues in clustering

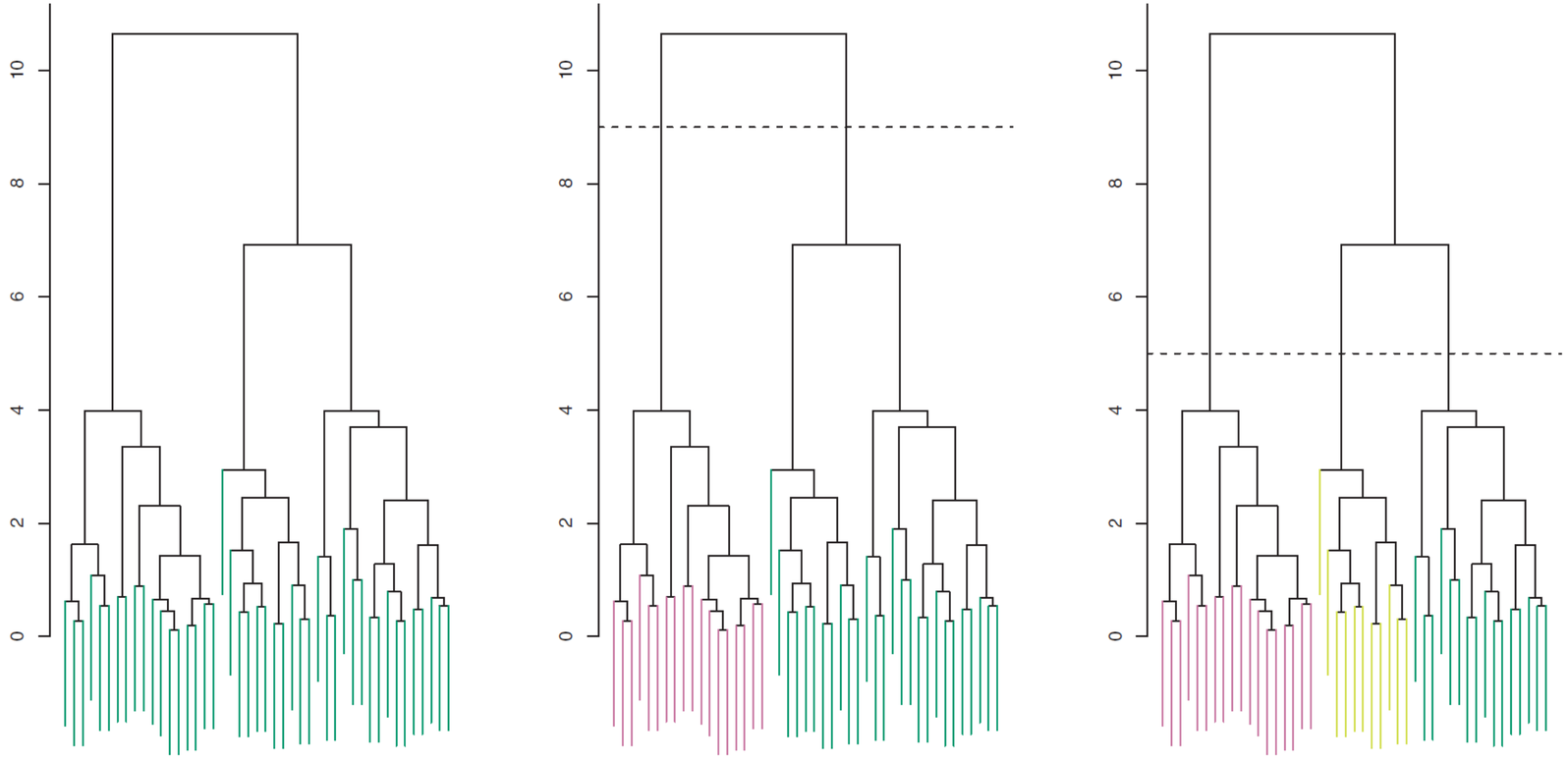
Hierarchical Clustering

- Hierarchical Clustering – Key Features:
  - Agglomerative (bottom-up) vs. Divisive (top-down)
  - Do NOT need to pre-specify the number of clusters
  - Generate a ***Dendrogram*** (upside-down tree)

- **<u>Hierarchical (Agglomerative) Clustering Algorithm</u>**
  1. Treat each of the n observations as its own cluster.
  2. For $i = n, n-1, \ldots, 2$:
     a) Among all i clusters, merge the two ***closest*** clusters.
     b) Compute the new pairwise inter-cluster distances among the $i-1$ remaining clusters.

Dendrogram

- Upside-down tree displaying the hierarchical clustering outcome
- *x*-axis: records
- *y*-axis: the inter-cluster distance where two clusters merge
- Cut the tree at different heights to yield different numbers of clusters



James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning: with applications in R.* Springer. 2nd Ed. p 523. Retrieved March 29, 2022 from, https://hastie.su.domains/ISLR2/ISLRv2_website.pdf.

Measuring Distances between Observations

- Euclidean distance
  - Most popular distance measure

$$d_{ij} = \sqrt{(x_{i1} - x_{j1})^2 + \cdots + (x_{ip} - x_{jp})^2}$$

- Correlation-based distance
  - Correlation between two *observations* based on all *variables*
  - Pearson correlation coefficient r_{ij}, $d_{ij} = 1 - r_{ij}^2$
- Other distance measures
  - Statistical distance
  - Manhattan distance

Distances between Clusters – Linkage (1/3)

- Single Linkage
  - The *smallest* distance among all pairwise distances between records in two clusters
  - Results in extended, trailing clusters (single records are merged one-at-a-time)
  - Clusters in elongated chain- or sausage-like shapes

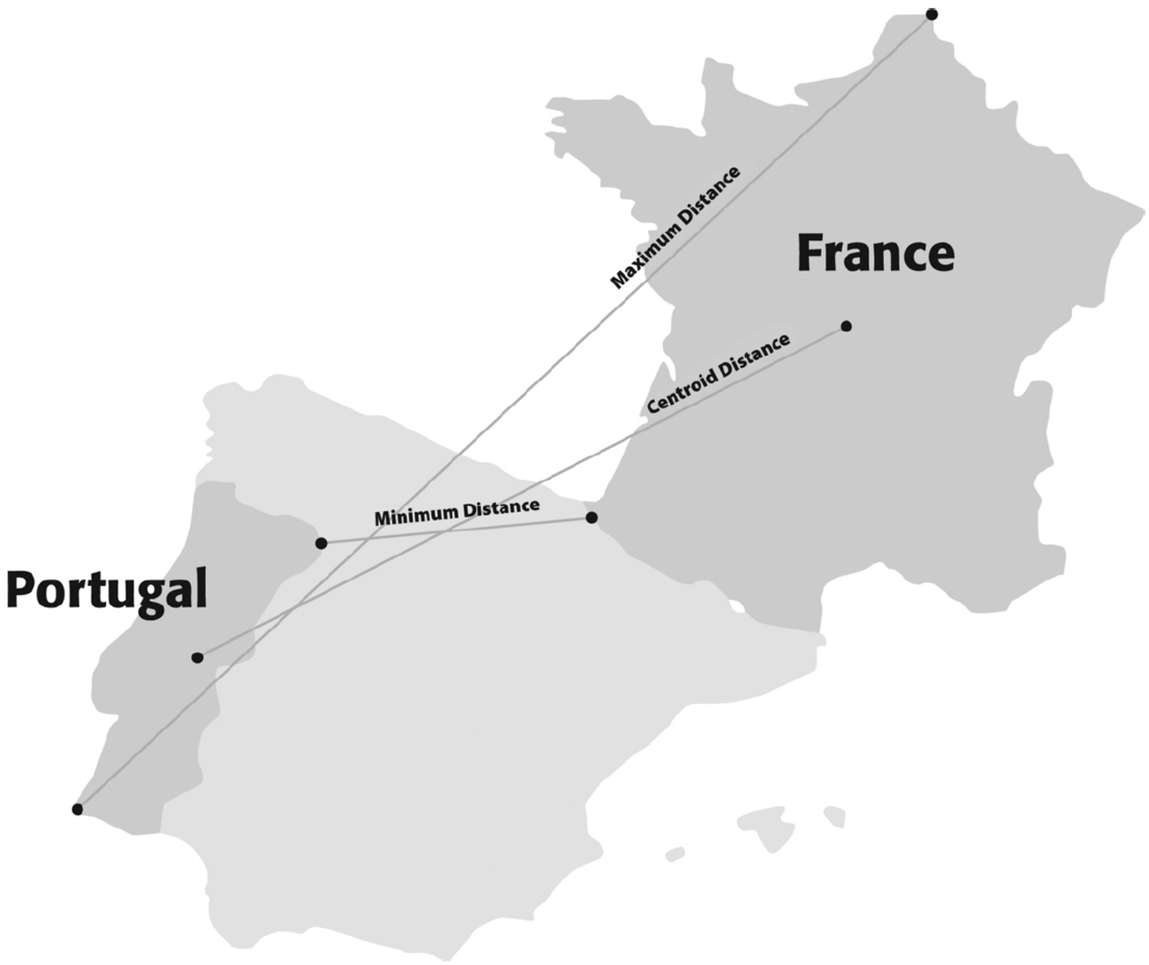
- Complete Linkage
  - The *largest* distance among all pairwise distances between records in two clusters
  - Clusters in spherical shapes
  - Single and complete linkages are based on *relative ordering* and <u>robust to changes of distance metric</u>

Distances between Clusters – Linkage (2/3)

- Average Linkage
  - The *average* of all pairwise distances between records in two clusters
  - More sensitive to changes of distance metric

- Centroid Linkage
  - The distance between the *centroids* of two clusters
  - ***Inversion***: two clusters merged at a height *below* either one in the dendrogram

Distances between Clusters – Linkage (3/3)



Shmueli, G., Bruce, P., Yahav, I., Patel, N. & Lichtendahl, K. (2017). *Data Mining for Business Analytics: Concepts, Techniques, and Applications in R*. John Wiley & Sons. p 367.

Limitations of Hierarchical Clustering

- Computationally expensive
  - Computing an $n \times n$ distance matrix
  - *K*-means less computationally intensive, preferred for large datasets
- Only one pass through the data
  - Records allocated incorrectly early in the process cannot be reallocated later
- Clusters obtained under different numbers of clusters are ***nested***
  - Some data may not be in such a nested structure (e.g., 2 gender groups, 3 income groups)
  - *K*-means are not necessarily nested, may perform better
- Low stability
  - Dropping a few records can lead to a different solution
  - Different linkages lead to different results

Practical Issues in Clustering

- Scale the variables
  - Centered to have mean zero (de-mean)
  - Scaled to have standard deviation one
- Choices of various parameters
  - Number of clusters; where to cut the dendrogram
  - Which distance metric; which linkage
- Validate cluster stability
  - Partition data into two parts, A and B; cluster part A
  - Assign each record in B to the nearest cluster centroid from A
  - Compare the cluster assignments to those from clustering all data

Practical Issues in Clustering (cont'd)

- Interpreting clusters
  - Obtain summary statistics (e.g., centroid) for each cluster
  - Use variables not used in clustering to examine cluster separation
  - Label the clusters

- *Taken with a grain of salt*
  - Not absolute truth, but a starting point for further analysis

Summary

- Hierarchical clustering algorithm
- Dendrograms
- Measure distances
  - Between observations
  - Between clusters
- Practical issues in clustering